

Area Under a Curve Using Limits of Sums

Date _____ Period _____

Evaluate each sum.

1) $\sum_{k=1}^n 18k$

2) $\sum_{k=1}^n (2k + 5)$

3) $\sum_{k=1}^n 8k^2$

4) $\sum_{k=1}^n (k^2 + 4)$

Evaluate each limit.

5) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{2}{n} + \frac{2k}{n^2} \right)$

6) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{5}{n} + \frac{k^3}{n^4} \right)$